



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 207



SUKI Banana Chipper

SUKI BANANA CHIPPER

Test Report No.	:	2013 - 207
Date of Test	:	June 12, 2013
Test Valid Until	:	June 12, 2018
Brand	:	SUKI
Model	:	-
Type	:	Flywheel-type, portable
Manufacturer	:	SUKI TRADING CORPORATION Agus, Ibabao Lapu-Lapu City
Test Requested by	:	Manufacturer
Machine Classification	:	Commercial

SUMMARY

The SUKI Banana Chipper driven by a 0.373 kW electric motor was tested using peeled unripe banana (commonly known as “saba”) with an average Moisture Content of 63.0%.

The SUKI Banana Chipper had an average Chipping Capacity of 204.5 kg/h banana at chipper shaft speed of 1781 rpm. The chips had an average thickness of 3.25 mm with thickness variation of $\pm 0.44\%$.

The average Power Consumption of the 0.373 kW electric motor was 0.32 kW with Load Current of 3.10 amperes and Line Voltage of 208 volts.

Two persons were needed to operate the machine: one for feeding and one for collecting the banana chips.

PURPOSE AND SCOPE OF TEST

The machine was tested for verification of its performance. The test was conducted at the plant site of the SUKI TRADING CORPORATION in Agus Ibabao, Lapu-Lapu City on June 12, 2013. The test was carried out to assess and verify the following:

1. Machine specifications
2. Performance
 - a. Chipping capacity
 - b. Speed of components
3. Power Consumption
4. Labor requirement
5. Quality of Work
 - a. Average thickness of chips
 - b. Variation in chips thickness

OPERATOR'S MANUAL

No Operator's Manual was submitted.

DESCRIPTION OF THE MACHINE

The SUKI Banana Chipper as shown in **Figure 1**, was a portable, flywheel-type chipping unit. It was powered by a 0.373 kW electric motor which directly drove the chipper's shaft through a coupler.

The machine consisted of an inlet chute and the chipping assembly. The inlet chute (**Fig. 2**) was where the bananas to be chipped were fed. It was made of 38 mm diameter, Schedule 20, stainless pipe.

The chipping assembly consisted of a chipping plate (**Fig. 3a**) with one piece of chipping blade (**Fig. 3b**). The chipping plate had dimensions (D x T) of 300 mm x 6 mm and made of stainless plate, while the blade had dimensions (L x W x T) of 75 mm x 50 mm x 1.5 mm and made of stainless plate. Below the chipping assembly was the outlet chute (**Fig. 4**) where banana chips flow down.

The whole frame of the machine was made of Schedule 20, hollow square tube.

Detailed specifications of the machine are found in **Table 1**.



Figure 1. SUKI Banana Chipper

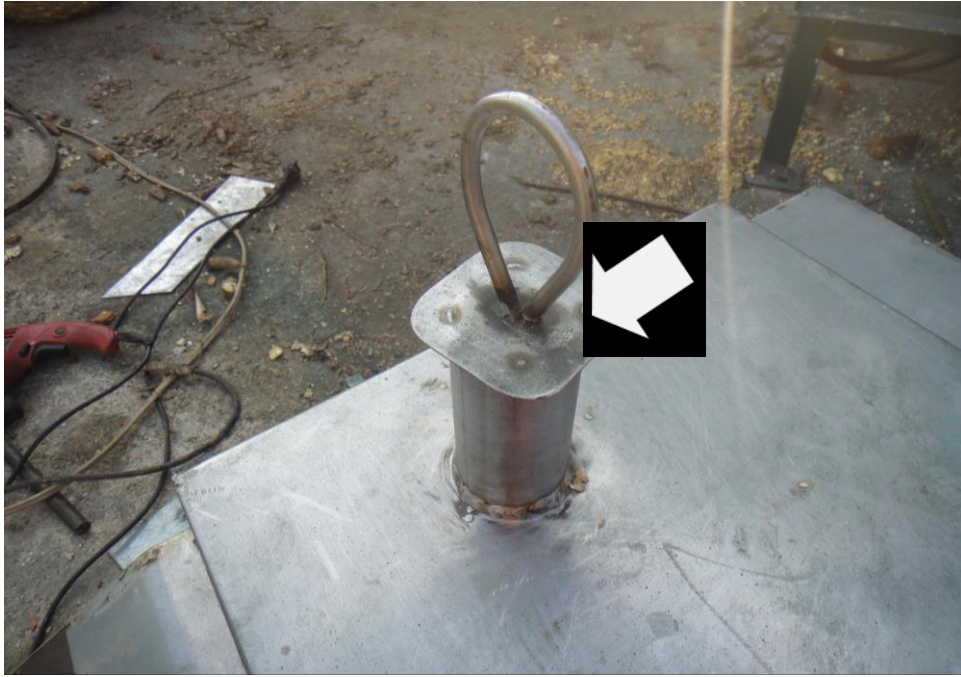


Figure 2. Inlet chute with cover

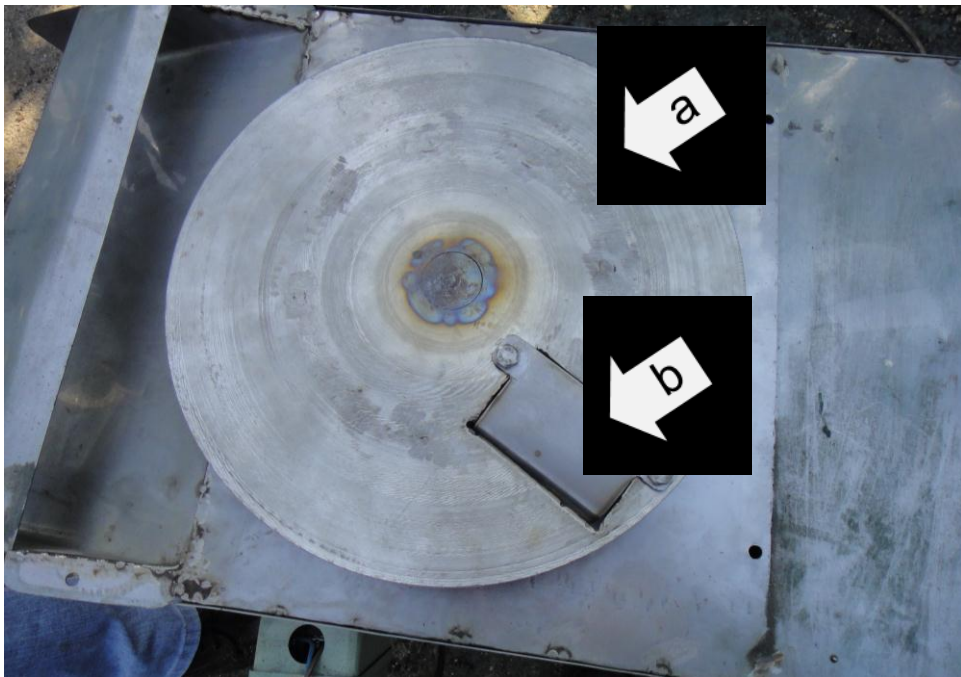


Figure 3. a) Chipper plate
b) Chipper blade

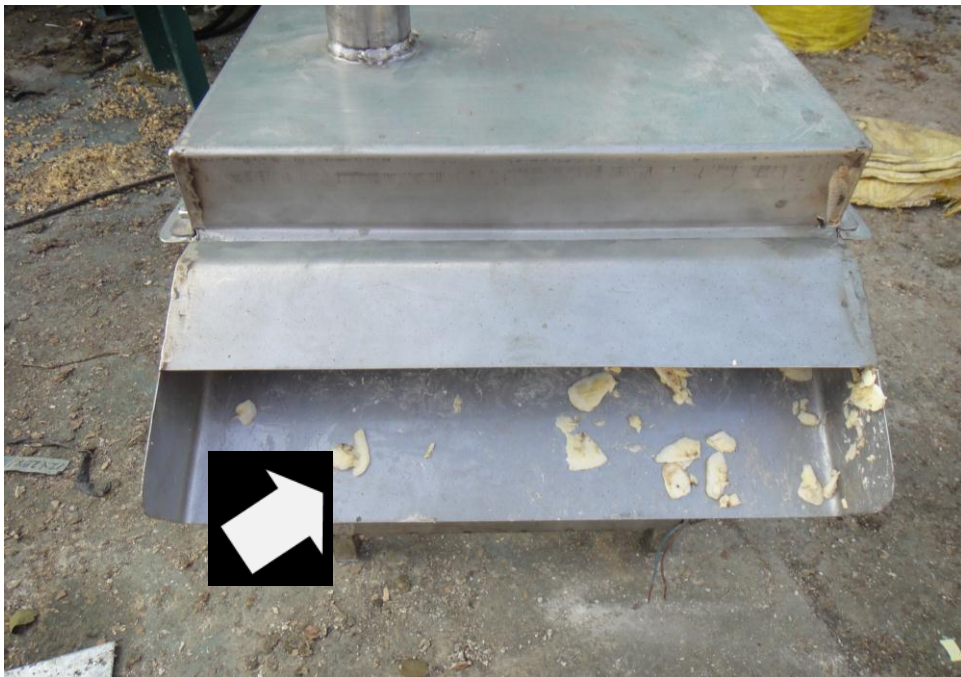


Figure 4. Outlet chute

SPECIFICATIONS

Table 1. Detailed specifications of the SUKI Banana Chipper

A.	Brand	:	SUKI
B.	Model	:	-
C.	Manufacturer	:	SUKI TRADING CORPORATION Agus Ibabao, Lapu-Lapu City
D.	Overall Dimensions and Weight		
1.	Length (mm)	:	610
2.	Width (mm)	:	410
3.	Height (mm)	:	690
4.	Weight (kg) without motor	:	No Data
E.	Inlet Chute		
1.	Size, L x D (mm)	:	160 x 200
2.	Height from the ground (mm)	:	605
3.	Material	:	Stainless Pipe, Sch. 20
F.	Chipping Assembly Cover		
1.	Material	:	Stainless Sheet, C-304 1.5 mm thick
G.	Chipper Plate		
1.	Size, D x T (mm)	:	300 x 6
2.	Material	:	Stainless Plate

Table 1. Detailed specifications of the SUKI Banana Chipper (cont'n.)

H.	Chipper Blade		
1.	No. of pieces	:	1
2.	Size, L x W x T (mm)	:	75 x 50 x 1.5
3.	Material	:	Stainless Plate, C-304
I.	Discharge Chute		
1.	Size, L x W (mm)	:	330 x 50
2.	Height from the ground (mm)	:	360
3.	Material	:	Stainless Sheet
J.	Main Frame		
1.	Material	:	Stainless Hollow Square Tube, Sch. 20
K.	Electric Motor (Prime mover)		
1.	Brand	:	POWER
2.	Type	:	Single-phase AC induction motor
3.	Make	:	People's Republic of China
4.	Rated power (kW)	:	0.373 kW
5.	Speed	:	1730
6.	Current (A)	:	4.2
7.	Voltage (V)	:	220
8.	Frequency (Hz)	:	60

RESULTS AND DISCUSSIONS

Results of performance tests are shown in **Table 2-3**.

Chipping Capacity

It refers to the amount of input material that can be chipped per unit time. Test results showed that using unripe peeled banana (commonly known as “saba”), the machine had a Chipping Capacity of 204.5 kg/h banana at blade shaft speed of 1781 rpm.

Quality of Work

The banana chips produced by the chipper had an average thickness of cut of 3.25 mm with Thickness Variation of ± 0.44 .

Power Consumption

The average Power Consumption of the 0.373 kW electric motor was 0.32 kW with Load Current of 3.10 amperes and Line Voltage of 208 volts.

Labor Requirement

Two (**Fig. 5**) persons were needed to operate the machine: one for feeding and one for collecting the banana chips.

Observation:

1. Unripe peeled “saba” banana (**Figure 6**) was used in the test. The chipper produced banana chips with average thickness of 3.25 mm (**Figure 7**).



Figure 5. Operators of the machine



Figure 6. Unripe peeled “saba” banana used in the test



Figure 7. Banana chips produced by the chipper

PERFORMANCE TEST

Machine Tested : SUKI Banana Chipper

Date of Test : June 12, 2013

Location of Test : Agus Ibabao, Lapu-Lapu City

Table 2. Conditions of Test Materials

ITEMS	Data
1. Crop	Banana
2. Variety	Unknown but commonly known as “saba”
3. Moisture Content, $MC_{w.b.}$ (%)	63.0
4. Dimension, L x D (mm)	85 x 36
5. Condition	Unripe and peeled

Table 3. Results of Performance Test

ITEMS	Data
1. Speed of Components (rpm)	
a) Motor/Chipper Shaft (rpm)	
Without load	1788
With load	1781
2. Power Consumption	
a) Power, kW	
Without load	0.30
With load	0.32
b) Voltage, V	
Without load	211
With load	208

Table 3. Results of Performance Test (cont'n.)

ITEMS	Data
c) Current, A Without load With load	3.0 3.1
3. Average thickness of cut, mm	3.25
4. Variation in thickness, %	± 0.44
5. Chipping Capacity, kg/h	204.5

Test Engineer:

DARWIN C. ARANGUREN

Checked by:

ROMULO E. EUSEBIO

Approved for Release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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